Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/22/2010
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Dose-rate effects of protons on the induction of genomic instability in vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Kanokporn.Rithidech@sbumed.org	**Fax:**	FY (631) 444-3424
PI Organization Type:	UNIVERSITY	**Phone:**	(631) 444-3446
Organization Name:	State University New York at Stony Brook		
PI Address 1:	Department of Pathology		
PI Address 2:	BHS T9 Health Sciences Center		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AP88G		
Performance Goal No.:			
Performance Goal Text:			
	Space environment consists of various types of radiation that are different from those found in the earth's atmosphere. Galactic cosmic rays (GCRs) and solar particle events (SPEs) are the greatest sources of radiation exposure to astronauts venturing beyond Earth. One of the major concerns associated with the exposure of the astronauts to radiation during space flights is cancer induction. Hence, it is vitally important to protect astronauts in space environments. This will require improvement in our knowledge of radiation-induced cellular and molecular changes that may impact the health of astronauts. Moreover, for research on radiation-induced genetic damage to be useful for risk assessment, it must be established in in vivo systems, especially in cells that are known to be at risk for health problems associated with radiation exposure such as bone marrow (BM) cells, the known target cell for radiation-induced leukemia. While the biological effects of GCR particles, in particular those with large mass (Z) and high energy (E) (known as HZE particles,		

Task Description:	such as 56Fe ions) have been investigated extensively, little is known about the biological effects of SPE components. The focus of this project is on 100 MeV protons because they are an abundant component of SPEs in space. During a journey to Mars, every cell in the body will be hit by a proton once every three days. However, little is known about early or late occurring in vivo biological effects of exposure to protons at doses and low dose rates that are similar to what astronauts encounter in space. We conducted a series of studies to fill this knowledge gap. In this project, we determine the type, the frequency, and the distribution of initial and subsequent chromosome aberrations, and the expression of genomic instability (assessed as late occurring chromosomal damage) induced in vivo in BM cells after exposure of mice to 100 MeV protons at the dose and dose rates encountered in space. Chromosomal damage is selected as an endpoint for the detection of proton-induced damage because of its strong correlation with the induction of cancers. The multi-color fluorescence in situ hybridization (mFISH) method is used for determining the types and the frequencies of chromosomal damage in BM cells of mice exposed to protons. Using the mFISH method, each individual pair of chromosomes (19 pairs of autosomes, X and Y chromosomes), and the genes they carry, is uniquely painted and simultaneously examined in the same cell. Classification of chromosomal damage can be done more precisely with the mFISH technology, compared to other cytogenetic assays. The mFISH method has never been used to study the biological effects of protons. Further, we also studied the effects of protons on the formation/reduction of phosphorylated histone 2AX (gamma H2AX), a biomarker for DNA double strand breaks, and on the cell cycle. Our results will deliver information that is of high priority for NASA in developing a better policy on safety limit and risk assessment associated with prolonged space flights. In addition to the above mentioned biological endpoints, our goal is to determine the potential molecular basis for the dose-rate dependencies of proton-induced genomic instability in vivo. To achieve this goal, we carefully measure levels of activated nuclear factor-kappa B (NF-kappa B), a highly important transcription factor (a protein that controls the transcription of the genes), and the expression of selected NF-kappa B target proteins involved in inflammatory response in BM cells collected at different times after irradiation from the same mice in which chromosomal damage is examined. Our research is the first to use integrative in vivo endpoints in the tissue (BM) that is also the target of radiation-induced myeloid leukemia to evaluate the dose-rate effects of protons. Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve the understanding of the molecular basis for dose rate dependencies of genomic instability. The results will provide high-priority information for NASA, because it has been widely suggested that elevation of genomic instability also elevates accumulation of genetic alterations that, in turn, increases cancer risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Upon completion of the in vivo work proposed in this study, our data will provide pivotal information that will lead to a better understanding of dose-rate effects and of mechanisms by which proton-induced damage can be potentially involved in the induction of genomic instability and cancer. The resulting data will deliver information that is of high priority for NASA. Our data will not only contribute to reducing the uncertainties in the assessment of cancer risk from exposure to protons during space flights but will also improve the understanding of the molecular basis for dose-rate dependencies of proton-induced genomic instability. Further, this study will couple molecular mechanisms with quantitative data from a robust and highly sensitive technique. Hence, the obtained data are critically important for understanding risks associated with space travel as well as developing advanced proton therapies for cancer treatment on earth. Further, knowing the pathways involved in response to radiation could be useful for planning countermeasure strategies to protect astronauts or space travelers in the space environment. Likewise, such knowledge can also have Earth benefits because a large number of people on Earth are exposed to radiation at varying doses and in many different ways. These include individuals exposed to either intentional sources (e.g. patients who receive low levels of radiation for medical diagnoses, patients who receive high doses of radiation from radiation therapy, victims of nuclear terrorism and members of the armed forces and astronauts) or accidental sources (e.g. workers in the nuclear power industry or people living in homes surrounding the power plants). It is therefore important to protect these populations at risk from radiation damage.
Task Progress:	We participated in several NSRL-BNL runs for proton-irradiation and completed the exposures. We exposed mice to 0, 0.5, and 1.0 Gy of 100 MeV protons (delivered at 5 or 10 mGy min-1). Samples have been collected from groups of mice (5 mice in each group) at various times post-irradiation and stored in suitable forms for analyses using the various endpoints mentioned above. We have made substantial discoveries during the past 12 months. Previously, we showed that a single dose of 1 Gy of 100 MeV protons produced similar effects (i.e. induction and repair of chromosomal damage, NF-kappa B activation, including expression of pro- and anti-inflammatory cytokines) to those induced by 1 Gy of 137Cs gamma rays when the same dose rate (10 mGy min-1) was used. Our new set of data clearly demonstrated that there is a dose-rate effect of 1 Gy of 100 MeV protons (delivered at 5 or 10 mGy min-1). We found that the level of chromosomal damage was higher with the dose rate of 10 mGy min-1 than that with the dose rate of 5 mGy min-1 at 3 or 24 hr post-irradiation. Our data also indicated that, regardless of the dose rate, there was a reduction in the frequencies of abnormal cells and chromosome aberrations, relative to those detected at 3 hr, suggesting a capacity of cells to repair the damage. The resulting data are the first to provide information on in vivo induction/repair of chromosome aberrations in BM cells of mice exposed to a single dose of 1 Gy of 100 MeV protons. Further, the gamma-H2AX data also indicate that damage and repair are dependent on the dose rate of 100 MeV protons. Taken together, our data provide evidence of more energy from a higher dose rate deposited leading to a higher level of chromosomal damage but lower ability to repair the damage. It will be important to determine whether chromosomal damage will persist at late time following exposure to 100 MeV protons. Such information will shed light on whether or not 100 MeV protons can induce genomic instability in vivo. Although the mFISH method is robust and highly sensitive, it is very time consuming and laborious. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide chromosome aberrations by mFISH will continue to be a major effort for year 4 of the study. We work diligently to generate data on the potential induction of genomic instability (measured by late-occurring chromosome aberrations) by protons for NASA.

	Our study also is the first in that the measurements of pro- and anti-inflammatory cytokine levels (at the protein level) are performed in BM cells of the same individual mouse after a whole body exposure to 100 MeV protons delivered at the dose rates comparable to those found in space. Our in vivo data demonstrated the impact of the dose rate of 100 MeV protons on the pattern of NF-kappa B activation (as well as the subsequent expression of proteins involved in inflammatory response). These are cytokines known to be regulated by activated NF-kappa B, i.e., tumor-necrosis factor-alpha (TNF-alpha), Interleukin-1 beta (IL-1 beta), and IL-6 in the total population of BM cells of exposed mice. We found highly significant levels of activated NF-kappa B, expression of pro-inflammatory cytokines TNF-alpha, IL-1beta, and IL-6 in BM cells of exposed mice at early time post-irradiation. These increases persisted up to 1 month post-irradiation. The detection of high levels of activated NF-kappaB and pro-inflammatory cytokines at late time (1 mo) post-exposure suggests the occurrence of the persistence of cytokine production, presumably due to feedback mechanisms. The balance or imbalance of the anti- and pro-inflammatory cytokines after radiation exposure may play an important role in determining the extent of the immune response and the health outcome. Our data provide an important initial basis for risk assessment of in vivo exposure to protons that are abundant in space during SPEs. Up to this point, our data indicate an in vivo dose-rate effect of 1 Gy of 100 MeV protons (delivered at 5 or 10 mGy min-1) on the key endpoints included in our study. We are currently examining the effects of a lower dose of 100 MeV protons (0.5 Gy) delivered at the same dose rates for 1 Gy. At the completion of this project, we will have a full insight into dose-rate effects of 0.5 and 1.0 Gy of 100 MeV protons delivered at 5 or 10 mGy min-1, the dose and dose-rate levels occurring in space environment.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Rithidech KN, Reungpatthanaphong P, Honikel L, Golightly M, Anderson CW. "Dose- rate effects of protons on the induction of genomic instability in vivo." Presented at the NASA Human Research Program, Houston, February 4-7, 2010. NASA Human Research Program, Houston, February 4-7, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Rithidech KN, Reungpatthanaphong P, Honikel L, Golightly M, Anderson CW. "Dose-rate effects of protons on the induction and repair of chromosome aberrations In Vivo." The 21st Annual Workshop NASA Space Radiation Health Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. The 21st Annual Workshop NASA Space Radiation Health Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Rithidech KN, Reungpatthanaphong P, Honikel L, Whorton EB. "Persistent activation of nuclear factor-kappa B and expression of pro-inflammatory cytokines in bone marrow cells after exposure of mice to protons." COSPAR 2010, the 38th Scientific Assembly of the Committee on Space Research, Bremen, Germany, July 2010. COSPAR 2010, the 38th Scientific Assembly of the Committee on Space Research, Bremen, Germany, July 2010. , Jul-2010
Abstracts for Journals and Proceedings	Reungpatthanaphong P, Honikel L, Golightly M, Anderson CW, Rithidech K. "Kinetics of in vivo formation and reduction of gamma H2AX levels in bone marrow cells after exposure of mice to 100 MeV/n protons." The Annual Meeting of Radiation Research Society, Maui, Hawaii, September 25-29, 2010. The Annual Meeting of Radiation Research Society, Maui, Hawaii, September 25-29, 2010. , Sep-2010
Articles in Peer-reviewed Journals	Rithidech KN, Supanpaiboon W, Honikel L, Whorton EB. "Induction of genomic instability after an acute whole-body exposure of mice to 56Fe ions." Adv Space Res, 2009 Oct 15;44(18):895-906. http://dx.doi.org/10.1016/j.asr.2009.05.001 , Oct-2009
Articles in Peer-reviewed Journals	Rithidech KN, Reungpatthanaphong P, Honikel L, Rusek A, Simon SR. "Dose-rate effects of protons on in vivo activation of nuclear factor-kappa B and cytokines in mouse bone marrow cells." Radiat Environ Biophys. 2010 May 28. [Epub ahead of print] PMID: 20508943 , May-2010